

1 UNCONDITIONAL LINEAR DENOISER WITH CIRCULANT MATRIX

A linear denoiser is $D(\mathbf{y}; \sigma) = \mathbf{W}_\sigma \mathbf{y} + \mathbf{b}_\sigma$, but now consider the case that $\mathbf{W}_\sigma$ is constrained to be circulant.

$$\mathcal{L}_\sigma^{\text{circ}} = \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} \|\mathbf{W}_\sigma^{\text{circ}}(\mathbf{x}_0 + \sigma \mathbf{Z}) + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2$$

where $\mathbf{W}_\sigma^{\text{circ}} \in \mathbb{R}^{d \times d}$, $\mathbf{x}_0 \in \mathbb{R}^{d \times 1}$. Diagonalize $\mathbf{W}_\sigma^{\text{circ}} := \mathbf{F}^* \text{diag}(\hat{\mathbf{W}}) \mathbf{F}$, where $\mathbf{F}$ is the unitary (across d) Discrete Fourier Transform matrix $\mathbf{F}$ where $F_{jk} = \frac{1}{\sqrt{d}} e^{-2\pi i j k / d}$ and $\mathbf{F}^*$ is the conjugate transpose where $F_{jk}^* = \frac{1}{\sqrt{d}} e^{2\pi i j k / d}$, and $\text{diag}(\hat{\mathbf{W}})$ is the Discrete Fourier Transform of the first row of the circulant matrix $\mathbf{W}$, i.e. $\text{diag}(\hat{\mathbf{W}})_j := \hat{w}_j = \sum_{k=0}^{d-1} w_k e^{-2\pi i j k / d}$ (w_k is the k -th entry of the first row of $\mathbf{W}$). For $\mathbf{W}$ to have real entries, $\hat{w}_j = \hat{w}_{d-j}$.

Recall that the optimal $\mathbf{b}_\sigma$ is $\mathbf{b}_\sigma^* = (\mathbf{I} - \mathbf{W}_\sigma) \mathbb{E}[\mathbf{x}_0]$. So we have:

$$\mathcal{L}_\sigma^{\text{circ}} = \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} \|\mathbf{W}_\sigma^{\text{circ}}(\mathbf{x}_0 - \mathbb{E}[\mathbf{x}_0]) + \sigma \mathbf{Z} - (\mathbf{x}_0 - \mathbb{E}[\mathbf{x}_0])\|_2^2$$

Let $\tilde{\mathbf{x}}_0 = \mathbf{x}_0 - \mathbb{E}[\mathbf{x}_0]$. Then:

$$\begin{aligned} \mathcal{L}_\sigma^{\text{circ}} &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I})} \|\mathbf{W}_\sigma^{\text{circ}}(\tilde{\mathbf{x}}_0 + \sigma \mathbf{Z}) - \tilde{\mathbf{x}}_0\|_2^2 \\ &= \text{Tr}(\mathbf{W}_\sigma^{\text{circ}}(\Sigma_{p_0} + \sigma^2 \mathbf{I}) \mathbf{W}_\sigma^{\text{circ} \top} - 2\mathbf{W}_\sigma^{\text{circ}} \Sigma_{p_0} + \Sigma_{p_0}) \end{aligned}$$

Let $\mathbf{P} = \mathbf{F} \Sigma_{p_0} \mathbf{F}^*$, then $\mathbf{P}$ is Hermitian, and $P_{kk} = f_k^H \Sigma_{p_0} f_k$ where f_k is the k -th column of $\mathbf{F}^*$, or the conjugate of the k -th row of $\mathbf{F}$. Note that $P_{kk} = P_{d-k, d-k}$. Since $\mathbf{W}_\sigma^{\text{circ}}$ is real valued, $\mathbf{W}_\sigma^{\text{circ} \top} = \mathbf{W}_\sigma^{\text{circ} H}$ ($\mathbf{A}^H$ is the conjugate transpose matrix of $\mathbf{A}$). So $\mathbf{W}_\sigma^{\text{circ} \top} = (\mathbf{F}^* \text{diag}(\hat{\mathbf{W}}) \mathbf{F})^H = \mathbf{F}^H \text{diag}(\hat{\mathbf{W}})^H (\mathbf{F}^*)^H = \mathbf{F}^* \text{diag}(\hat{\mathbf{W}})^H \mathbf{F}$.

Since $\mathbf{F}$ and $\mathbf{F}^*$ are both symmetric, $\mathbf{W}_\sigma^{\text{circ} \top} = \mathbf{F}^* \text{diag}(\hat{\mathbf{W}})^H \mathbf{F}$ is also symmetric. Consider the first term:

$$\begin{aligned} &\text{Tr}(\mathbf{F}^* \text{diag}(\hat{\mathbf{W}}) \mathbf{F} (\Sigma_{p_0} + \sigma^2 \mathbf{I}) \mathbf{F}^* \text{diag}(\hat{\mathbf{W}})^H \mathbf{F}) \\ &= \text{Tr}(\mathbf{F} \mathbf{F}^* \text{diag}(\hat{\mathbf{W}}) \mathbf{F} (\Sigma_{p_0} + \sigma^2 \mathbf{I}) \mathbf{F}^* \text{diag}(\hat{\mathbf{W}})^H) \\ &= \text{Tr}(\text{diag}(\hat{\mathbf{W}}) (\mathbf{P} + \sigma^2 \mathbf{I}) \text{diag}(\hat{\mathbf{W}})^H) \\ &= \sum_{k=1}^d \|\hat{w}_k\|^2 (P_{kk} + \sigma^2) \end{aligned}$$

Consider the second term:

$$\begin{aligned} &\text{Tr}(\mathbf{F}^* \text{diag}(\hat{\mathbf{W}}) \mathbf{F} \Sigma_{p_0}) \\ &= \text{Tr}(\text{diag}(\hat{\mathbf{W}}) \mathbf{F} \Sigma_{p_0} \mathbf{F}^*) \\ &= \text{Tr}(\text{diag}(\hat{\mathbf{W}}) \mathbf{P}) = \sum_{k=1}^d \hat{w}_k P_{kk} \end{aligned}$$

Since $P_{kk} = P_{d-k, d-k}$, pairing k and $d-k$ in this sum gives us: $\hat{w}_k P_{kk} + \hat{w}_{d-k} P_{d-k, d-k} = P_{kk} (\hat{w}_k + \bar{\hat{w}}_k) = P_{kk} 2\text{Re}(\hat{w}_k)$. So $\text{Tr}(\mathbf{F}^* \text{diag}(\hat{\mathbf{W}}) \mathbf{F} \Sigma_{p_0}) = \sum_{k=1}^d \text{Re}(\hat{w}_k) P_{kk}$. Then:

$$\mathcal{L}_\sigma^{\text{circ}} = \sum_{k=1}^d (\|\hat{w}_k\|^2 (P_{kk} + \sigma^2) - 2\text{Re}(\hat{w}_k) P_{kk}) + \text{Tr}(\Sigma_{p_0})$$

Now observe that each $\hat{w}_k$ is independent and so we can optimize for each $\hat{w}_k$ independently. Let each $\hat{w}_k = a + bi$ where $a, b \in \mathbb{R}$. Our objective is to minimize:

$$\|\hat{w}_k\|^2 (P_{kk} + \sigma^2) - 2\text{Re}(\hat{w}_k) P_{kk} = (a^2 + b^2) (P_{kk} + \sigma^2) - 2a P_{kk}$$

Note that P_{kk} is real and non-negative, so to achieve the minimum of $\mathcal{L}_\sigma^{\text{circ}}$, it ought to be that $b = 0$ and solving for the minimum of w.r.t a :

$$\nabla_a a^2 (P_{kk} + \sigma^2) - 2a P_{kk} = 2a (P_{kk} + \sigma^2) - 2P_{kk} = 0$$

so:

$$a = \frac{P_{kk}}{P_{kk} + \sigma^2} = \hat{w}_k^*$$

Therefore:

$$\mathcal{L}_\sigma^{\text{circ}} = \sum_{k=1}^d \frac{-P_{kk}^2}{P_{kk} + \sigma^2} + \text{Tr}(\Sigma_{p_0})$$

Note that $\sum_{k=1}^d P_{kk} = \sum_{k=1}^d \text{Tr}(f_k^H \Sigma_{p_0} f_k) = \text{Tr}(\Sigma_{p_0} \sum_{k=1}^d f_k f_k^H) = \text{Tr}(\Sigma_{p_0} \mathbf{F} \mathbf{F}^*) = \text{Tr}(\Sigma_{p_0} \mathbf{I}) = \text{Tr}(\Sigma_{p_0})$, so:

$$\mathcal{L}_\sigma^{\text{circ}} = \sum_{k=1}^d \frac{-P_{kk}^2}{P_{kk} + \sigma^2} - P_{kk} = \sigma^2 \sum_{k=1}^d \frac{P_{kk}}{P_{kk} + \sigma^2}$$

Recall that the regular optimal denoiser loss is:

$$\mathcal{L}_\sigma = \sigma^2 \sum_{k=1}^d \frac{\lambda_k}{\lambda_k + \sigma^2}$$

where λ_k is the eigenvalues of Σ_{p_0} . Since $\mathbf{F}\mathbf{F}^* = \mathbf{I}$, the transformation $\mathbf{P} = \mathbf{F}\Sigma_{x_0}\mathbf{F}^*$ preserves the eigenvalues of Σ_{x_0} . By the Schur-Horn Theorem, for any Hermitian matrix, the vector of its true eigenvalues always majorizes the vector of its diagonal entries, i.e. $\lambda \succ \text{diag}(\mathbf{P})$. Let $f(x) = \frac{x}{x+\sigma^2}$; it can be confirmed that $f(x)$ is concave since the second derivative $f''(x) = \frac{-2\sigma^2}{(x+\sigma^2)^3}$ is strictly negative for $x \geq 0$ (note that λ_k and $\mathbf{P}_{kk}$ are non-negative). By the Karamata's Inequality/Majorization Theorem, if a vector $\mathbf{x}$ majorizes $\mathbf{y}$ and f is concave, then $\sum_{k=1}^d f(x_k) \leq \sum_{k=1}^d f(y_k)$. So:

$$\mathcal{L}_\sigma^{\text{circ}} = \sigma^2 \sum_{k=1}^d \frac{\mathbf{P}_{kk}}{\mathbf{P}_{kk} + \sigma^2} \geq \sigma^2 \sum_{k=1}^d \frac{\lambda_k}{\lambda_k + \sigma^2} = \mathcal{L}_\sigma$$

Constraint for $\mathbf{W}_\sigma$ to be circulant doesn't buy us anything in the linear denoiser case; since circulant matrices is still a subset of matrices and so the optimal denoiser loss ought to be the lower bound for the circulant matrix denoiser loss.

2 CONDITIONAL LINEAR DENOISER WITH CIRCULANT MATRIX

Now similarly consider the conditional denoiser case, where we also constrain $\mathbf{W}_\sigma$ to be circulant. Recall the optimal $\mathbf{b}_\sigma$ is $\mathbf{b}_\sigma^* = (\mathbf{I} - \mathbf{W}_\sigma)\mu_{p_0} - \mathbf{V}_\sigma\mu_U$ and $\mathbf{V}_\sigma^* = (\mathbf{I} - \mathbf{W}_\sigma)\text{Cov}(\mathbf{x}_0, \mathbf{U})\Sigma_U^{-1}$

$$\begin{aligned} \mathcal{L}_{\sigma, U}^{\text{circ}} &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I}), \mathbf{U}} \|\mathbf{W}_\sigma^{\text{circ}}(\mathbf{x}_0 + \sigma\mathbf{Z}) + \mathbf{V}_\sigma\mathbf{U} + \mathbf{b}_\sigma - \mathbf{x}_0\|_2^2 \\ &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I}), \mathbf{U}} \|\mathbf{W}_\sigma^{\text{circ}}(\mathbf{x}_0 + \sigma\mathbf{Z}) + \mathbf{V}_\sigma\mathbf{U} + (\mathbf{I} - \mathbf{W}_\sigma^{\text{circ}})\mu_{p_0} - \mathbf{V}_\sigma\mu_U - \mathbf{x}_0\|_2^2 \\ &= \mathbb{E}_{\mathbf{x}_0 \sim p_0, \mathbf{Z} \sim \mathcal{N}(0, \mathbf{I}), \mathbf{U}} \|\mathbf{W}_\sigma^{\text{circ}}(\tilde{\mathbf{x}}_0 + \sigma\mathbf{Z}) - \tilde{\mathbf{x}}_0 + \mathbf{V}_\sigma\tilde{\mathbf{U}}\|_2^2 \\ &= \text{Tr}(\mathbf{W}_\sigma^{\text{circ}}(\Sigma_{p_0} + \sigma^2\mathbf{I})\mathbf{W}_\sigma^{\text{circ}\top} - 2\mathbf{W}_\sigma^{\text{circ}}\Sigma_{p_0} + \Sigma_{p_0} + \mathbf{V}_\sigma\Sigma_U\mathbf{V}_\sigma^\top + 2\mathbf{V}_\sigma\text{Cov}(\mathbf{U}, \mathbf{x}_0)(\mathbf{W}_\sigma^{\text{circ}} - \mathbf{I})) \\ &= \text{Tr}(\mathbf{W}_\sigma^{\text{circ}}(\Sigma_{p_0} + \sigma^2\mathbf{I})\mathbf{W}_\sigma^{\text{circ}\top} - 2\mathbf{W}_\sigma^{\text{circ}}\Sigma_{p_0} + \Sigma_{p_0} - (\mathbf{I} - \mathbf{W}_\sigma)\text{Cov}(\mathbf{x}_0, \mathbf{U})\Sigma_U^{-1}\text{Cov}(\mathbf{U}, \mathbf{x}_0)(\mathbf{I} - \mathbf{W}_\sigma)^\top) \\ &= \text{Tr}(\mathbf{W}_\sigma^{\text{circ}}(\Sigma_{p_0} + \sigma^2\mathbf{I} - \text{Cov}(\mathbf{x}_0, \mathbf{U})\Sigma_U^{-1}\text{Cov}(\mathbf{U}, \mathbf{x}_0))\mathbf{W}_\sigma^{\text{circ}\top} - 2\mathbf{W}_\sigma^{\text{circ}}(\Sigma_{p_0} - \text{Cov}(\mathbf{x}_0, \mathbf{U})\Sigma_U^{-1}\text{Cov}(\mathbf{U}, \mathbf{x}_0)) \\ &\quad + (\Sigma_{p_0} - \text{Cov}(\mathbf{x}_0, \mathbf{U})\Sigma_U^{-1}\text{Cov}(\mathbf{U}, \mathbf{x}_0))) \end{aligned}$$

Let $\Sigma_{\text{cond}} = \Sigma_{p_0} - \text{Cov}(\mathbf{x}_0, \mathbf{U})\Sigma_U^{-1}\text{Cov}(\mathbf{U}, \mathbf{x}_0)$, then:

$$\mathcal{L}_{\sigma, U}^{\text{circ}} = \text{Tr}(\mathbf{W}_\sigma^{\text{circ}}(\Sigma_{\text{cond}} + \sigma^2\mathbf{I})\mathbf{W}_\sigma^{\text{circ}\top} - 2\mathbf{W}_\sigma^{\text{circ}}\Sigma_{\text{cond}} + \Sigma_{\text{cond}})$$

This is the same form as $\mathcal{L}_\sigma^{\text{circ}}$ except with Σ_{p_0} swapped out for Σ_{cond} . Hence we can draw the same conclusion: let $\mathbf{P}^{\text{cond}} = \mathbf{F}\Sigma_{\text{cond}}\mathbf{F}^*$ and as before, $\text{diag}(\hat{\mathbf{W}})$ is the Discrete Fourier Transform of the first row of the circulant matrix $\mathbf{W}$, i.e. $\text{diag}(\hat{\mathbf{W}})_j := \hat{w}_j = \sum_{k=0}^d w_k e^{-2\pi i j k / d}$ (w_k is the k -th entry of the first row of $\mathbf{W}$). For $\mathbf{W}$ to have real entries, $\hat{w}_j = \hat{w}_{d-j}$. Let $\mathbf{P}_{kk}^{\text{cond}} = f_k^H \Sigma_{\text{cond}} f_k$. Then:

$$\mathcal{L}_{\sigma, U}^{\text{circ}} = \sum_{k=1}^d (\|\hat{w}_k\|^2 (\mathbf{P}_{kk}^{\text{cond}} + \sigma^2) - 2\text{Re}(\hat{w}_k) \mathbf{P}_{kk}^{\text{cond}}) + \text{Tr}(\Sigma_{\text{cond}})$$

Recall that $\Sigma_{\text{cond}} = \Sigma_{p_0} - \text{Cov}(\mathbf{x}_0, \mathbf{U})\Sigma_U^{-1}\text{Cov}(\mathbf{U}, \mathbf{x}_0) = \text{Var}(\mathbf{x}_0 | \mathbf{U})$ so Σ_{cond} is positive semi-definite and so $\mathbf{P}_{kk}^{\text{cond}}$ is real and non-negative. Similar to above, optimize for each $\hat{w}_k$ independently. Let $\hat{w}_k = a + bi$ where $a, b \in \mathbb{R}$; we're minimizing:

$$\|\hat{w}_k\|^2 (\mathbf{P}_{kk}^{\text{cond}} + \sigma^2) - 2\text{Re}(\hat{w}_k) \mathbf{P}_{kk}^{\text{cond}} = (a^2 + b^2)(\mathbf{P}_{kk}^{\text{cond}} + \sigma^2) - 2a\mathbf{P}_{kk}^{\text{cond}}$$

Again it ought to be that $b = 0$ and:

$$a = \frac{\mathbf{P}_{kk}^{\text{cond}}}{\mathbf{P}_{kk}^{\text{cond}} + \sigma^2} = \hat{w}_k^*$$

And so:

$$\mathcal{L}_{\sigma, U}^{\text{circ}} = \sigma^2 \sum_{k=1}^d \frac{\mathbf{P}_{kk}^{\text{cond}}}{\mathbf{P}_{kk}^{\text{cond}} + \sigma^2}$$

Recall that the optimal conditional linear denoiser loss without any constraint on $\mathbf{W}_\sigma$ is:

$$\mathcal{L}_{\sigma, U} = \sigma^2 \sum_{k=1}^d \frac{\lambda_k^{\text{cond}}}{\lambda_k^{\text{cond}} + \sigma^2}$$

where λ_k^{cond} is the eigenvalues of Σ_{cond} , and also the eigenvalues of $\mathbf{P}^{\text{cond}}$. Then like before, by the Schur-Horn Theorem, for the Hermitian $\mathbf{P}^{\text{cond}}$, $\lambda^{\text{cond}} \succ \text{diag}(\mathbf{P}^{\text{cond}})$. And so by the Karamata's Inequality/Majorization Theorem:

$$\mathcal{L}_{\sigma, U}^{\text{circ}} = \sigma^2 \sum_{k=1}^d \frac{\mathbf{P}_{kk}^{\text{cond}}}{\mathbf{P}_{kk}^{\text{cond}} + \sigma^2} \geq \sigma^2 \sum_{k=1}^d \frac{\lambda_k^{\text{cond}}}{\lambda_k^{\text{cond}} + \sigma^2} = \mathcal{L}_{\sigma, U}$$

Again in the conditional case, constraining $\mathbf{W}_\sigma$ to be circulant doesn't buy us anything more than the optimal linear denoiser does. Also note that the gain from going from unconditional to conditional in the case of circulant $\mathbf{W}_\sigma$ is the gain of going from $\text{Var}(\mathbf{x}_0) = \Sigma_{p_0}$ to $\text{Var}(\mathbf{x}_0 | U) = \Sigma_{\text{cond}}$ w.r.t. getting the values of $\mathbf{P}_{kk} = f_k^H \Sigma_{p_0} f_k$ v.s. $\mathbf{P}_{kk}^{\text{cond}} = f_k^H \Sigma_{\text{cond}} f_k$.

3 MUTUAL INFORMATION OF THE LEARNED DISTRIBUTION

Recall that with parametrization $\gamma = 1/t$, and $\gamma = 1/\sigma^2$ then mutual information is:

$$I(\mathbf{x}_0; Y_\gamma) = \frac{1}{2} \int_0^\gamma \text{MMSE}[\mathbf{x}_0 | Y_s] ds$$

Consider the affine circulant function class:

$$\mathcal{F}_{\text{circ}} = \{y \mapsto \mathbf{W}_\sigma y + b : \mathbf{W}_\sigma \text{ is circulant}\}$$

So the constrained MMSE by Tweedie's formula is:

$$\text{MMSE}_{\mathcal{F}_{\text{circ}}}[\mathbf{x}_0 | Y_\gamma] = \min_{\mathbf{W}_{\text{circulant}}, b} \mathbb{E} \|\mathbf{x}_0 - (\mathbf{W}_\sigma Y_\gamma + b)\|_2^2$$

From last section, with $\gamma = 1/\sigma^2$, then:

$$\text{MMSE}_{\mathcal{F}_{\text{circ}}}[\mathbf{x}_0 | Y_\gamma] = \sum_{k=1}^d \frac{\mathbf{P}_{kk}}{1 + \gamma \mathbf{P}_{kk}}$$

Then the constrained mutual information is:

$$\begin{aligned} I_{\mathcal{F}_{\text{circ}}}(\mathbf{x}_0; Y_\gamma) &= \frac{1}{2} \int_0^\gamma \sum_{k=1}^d \frac{\mathbf{P}_{kk}}{1 + s \mathbf{P}_{kk}} ds \\ &= \frac{1}{2} \sum_{k=1}^d \log(1 + \gamma \mathbf{P}_{kk}) \end{aligned}$$

Recall that in $\mathcal{F}_L$ the linear/affine space (unconstrained) $\mathcal{F}_L = \{y \mapsto \mathbf{W}_\sigma y + b\}$, we have:

$$I_{\mathcal{F}_L}(\mathbf{x}_0; Y_\gamma) = \frac{1}{2} \sum_{k=1}^d \log(1 + \gamma \lambda_k)$$

where λ_k is the eigenvalues of Σ_{p_0} . Again we know that $\boldsymbol{\lambda} \succ \text{diag}(\mathbf{P})$, and since $f(x) = \log(1 + \gamma x)$ is concave, by the Karamata's Inequality/Majorization Theorem, $\sum_{k=1}^d f(\lambda_k) \leq \sum_{k=1}^d f(\mathbf{P}_{kk})$. Therefore:

$$I_{p_0}(\mathbf{x}_0; Y_\gamma) \leq I_{\mathcal{F}_L}(\mathbf{x}_0; Y_\gamma) \leq I_{\mathcal{F}_{\text{circ}}}(\mathbf{x}_0; Y_\gamma)$$

The second inequality also makes sense because

On the other hand, entropy is:

$$h(Y_\gamma | \mathbf{x}_0) = \frac{d}{2} \log \left(\frac{2\pi e}{\gamma} \right)$$

and

$$h(Y_\gamma) = \frac{d}{2} \log \left(\frac{2\pi e}{\gamma} \right) + I(\mathbf{x}_0; Y_\gamma)$$

Therefore:

$$\begin{aligned} h(Y_\gamma) &\leq \frac{d}{2} \log \left(\frac{2\pi e}{\gamma} \right) + I_{\mathcal{F}_{\text{circ}}}(\mathbf{x}_0; Y_\gamma) \\ &= \frac{1}{2} \sum_{k=1}^d \log \left(2\pi e \left(\mathbf{P}_{kk} + \frac{1}{\gamma} \right) \right) \end{aligned}$$

Again since $\gamma = 1/\sigma^2$:

$$h(\mathbf{x}_0 + \sigma \mathbf{Z}) \leq \frac{1}{2} \log (2\pi e (\mathbf{P}_{kk} + \sigma^2))$$

For the conditional case:

$$\text{MMSE}_{\mathcal{F}_{\text{circ}}, U}[\mathbf{x}_0 | Y_\gamma, U] = \sum_{k=1}^d \frac{\mathbf{P}_{kk}^{\text{cond}}}{1 + \gamma \mathbf{P}_{kk}^{\text{cond}}}$$

So:

$$\begin{aligned} I_{\mathcal{F}_{\text{circ}}}(Y_\gamma; U) &= \frac{1}{2} \int_0^\gamma \sum_{k=1}^d \text{MMSE}_{\mathcal{F}_{\text{circ}}}[\mathbf{x}_0 | Y_\gamma] - \text{MMSE}_{\mathcal{F}_{\text{circ}}, U}[\mathbf{x}_0 | Y_\gamma, U] ds \\ &= \frac{1}{2} \sum_{k=1}^d \log \left(\frac{1 + \gamma \mathbf{P}_{kk}}{1 + \gamma \mathbf{P}_{kk}^{\text{cond}}} \right) \end{aligned}$$

Taking $\gamma \rightarrow \infty$:

$$I_{\mathcal{F}_{\text{circ}}}(\mathbf{x}_0; U) = \frac{1}{2} \sum_{k=1}^d \log \left(\frac{\mathbf{P}_{kk}}{\mathbf{P}_{kk}^{\text{cond}}} \right)$$

Recall that in $\mathcal{F}_L$ the unconstrained linear/affine space:

$$I_{\mathcal{F}_L}(\mathbf{x}_0; U) = \frac{1}{2} \sum_{k=1}^d \log \left(\frac{\lambda_k}{\lambda_k^{\text{cond}}} \right)$$

Unlike the unconditional case we can't really compare $I_{\mathcal{F}_{\text{circ}}}(\mathbf{x}_0; U)$ and $I_{\mathcal{F}_L}(\mathbf{x}_0; U)$.

We can see that the circulant constraint increases its function class MMSE and so increases the corresponding entropy upperbound.

4 UNCONDITIONAL NONLINEAR DENOISER WITH CIRCULANT MATRIX

Recall that the nonlinear map is:

$$\phi(\mathbf{y}) = \omega(\Theta \mathbf{y} + \epsilon)$$

where $\Theta \in \mathbb{R}^{k \times d}$ and $\epsilon \in \mathbb{R}^k$ are random matrices and the denoiser w.r.t this non-linear feature map is:

$$D(\mathbf{y}; \sigma) = \mathbf{W}_\sigma \phi(\mathbf{y}) + \mathbf{b}_\sigma$$

where $\mathbf{W}_\sigma \in \mathbb{R}^{d \times k}$ and $\mathbf{b}_\sigma \in \mathbb{R}^d$. Since circulant matrices need to be square, assuming k is divisible by d (which is the setup of our RF scaling experiment where we drive k/d to ∞), we can constrain Θ and $\mathbf{W}_\sigma$ to be circulant by block of $d \times d$. Let $k = c \cdot d$, write $\Theta = [\Theta_1 \ \Theta_2 \ \dots \ \Theta_c]^\top$ where each Θ_a is a $d \times d$ circulant matrix. Correspondingly, $\mathbf{W}_\sigma = [\mathbf{W}_1 \ \mathbf{W}_2 \ \dots \ \mathbf{W}_c]$, each $\mathbf{W}_a$ for $a \in \{1, 2, \dots, c\}$ is $d \times d$ circulant. Now, a circulant matrix is determined by the entries of its first row, and the following rows after is just a cyclic shift (each by one entry to the right), for example:

$$\Theta_a = \begin{bmatrix} a_1 & a_2 & \dots & a_d \\ a_d & a_1 & a_2 & \dots \\ \vdots & \vdots & \vdots & \vdots \\ a_2 & \dots & a_d & a_1 \end{bmatrix}$$

Denote this cyclic shift as $\Theta_a = \text{circ}(h_a)$ where $h_a = [a_1 \ a_2 \ \dots \ a_d]$. Under the random-feature framework, $h_a \sim \mathcal{N}(0, \frac{1}{d} \mathbf{I})$. Hence each entry in the first row and so in the entire matrix has marginal distribution $\mathcal{N}(0, 1/d)$, the same as in the regular case where there is no circulant constraint on Θ and $\mathbf{W}_\sigma$. Our RF matrices in this case differ only in the joint distribution, where each row in Θ_a is completely dependent on the first row and each other; while in the regular case the rows are independent. We have:

$$\Theta \mathbf{y} = [\Theta_1 \mathbf{y} \ \Theta_2 \mathbf{y} \ \dots \ \Theta_c \mathbf{y}]^\top$$

Let's index the rows if Θ with $\{j = (a, \tau)\}$ where $a \in 1, 2, \dots, c$ is indexing into the circulant blocks (there are c blocks) and $\tau = \{1, 2, \dots, d-1\}$ is indexing into the rows of a circulant block starting from after the first rows; each row is cyclic shift by τ entry to the right of the first row $h_a = [a_1 \ a_2 \ \dots \ a_d]$, i.e. $\Theta_{(a, \tau)} = S^\tau h_a$, for example $\Theta_{(a, 1)} = S^1 h_a = [a_d \ a_1 \ a_2 \ \dots]$, $\Theta_{(a, d-1)} = S^{d-1} h_a = [a_2 \ \dots \ a_d \ a_1]$

S is the shift operator where $(Sx)_i = x_{i-1 \bmod d}$, so $(S^\tau x)_i = x_{i-\tau}$, and $S^\top = S^{-1}$.

For now let the noise ϵ be the same for each row in a block; i.e. $\epsilon_{(a, \tau)} = \epsilon_a$, so noise is the same for each feature in the same circulant block; since each row is a cyclic shift of the first, this is to avoid random noise diluting the effect of the cyclic shift (intuitively?, I'll come back to this later). So ϵ is drawn once and then fixed for each block. We have:

$$[\Theta_a \mathbf{y}]_\tau = \Theta_{(a, \tau)}^\top \mathbf{y} = h_a^\top (S^\tau)^\top \mathbf{y} = h_a^\top S^{-\tau} \mathbf{y}$$

Let $\phi_a(\mathbf{y}) = \omega(\Theta_a \mathbf{y} + \epsilon_a)$, then:

$$D(\mathbf{y}; \sigma) = \sum_{a=1}^c \mathbf{W}_a \phi_a(\mathbf{y}) + \mathbf{b}_\sigma$$

Let each $\mathbf{W}_a = \mathbf{F}^* \text{diag}(\hat{\mathbf{W}}_a) \mathbf{F}$, where $\text{diag}(\hat{\mathbf{W}}_a)_j := \hat{w}_j^a = \sum_{k=0}^d w_k^a e^{-2\pi i j k / d}$ (w_k^a is the k -th entry of the first row of $\mathbf{W}_a$). Recall from last write-up the denoiser loss is:

$$\mathcal{L}_\sigma^{\text{circ}} = \text{Tr}(\mathbf{W}_\sigma \Sigma_\phi \mathbf{W}_\sigma^\top - 2\mathbf{W}_\sigma \Sigma_{\phi, \mathbf{x}_0} + \Sigma_{\mathbf{p}_0})$$

Σ_ϕ is $cd \times cd$, we can partition into c^2 blocks of $d \times d$, denote each block $\Sigma_\phi^{(a,b)} = \Sigma_{\phi_a, \phi_b}$ where $1 \leq a, b \leq c$. $\Sigma_{\phi, \mathbf{x}_0}$ is $cd \times d$, so we can partition into c blocks of $d \times d$ along its rows, denote each block $\Sigma_{\phi_a, \mathbf{x}_0}$. Consider:

$$\mathbf{W}_\sigma \Sigma_\phi = [\sum_a \mathbf{W}_a \Sigma_{\phi_a, \phi_1} \quad \sum_a \mathbf{W}_a \Sigma_{\phi_a, \phi_2} \quad \cdots \quad \sum_a \mathbf{W}_a \Sigma_{\phi_a, \phi_c}]$$

So:

$$\begin{aligned} \mathbf{W}_\sigma \Sigma_\phi \mathbf{W}_\sigma^\top &= \sum_a \mathbf{W}_a \Sigma_{\phi_a, \phi_1} \mathbf{W}_1^\top + \sum_a \mathbf{W}_a \Sigma_{\phi_a, \phi_2} \mathbf{W}_2^\top + \cdots + \sum_a \mathbf{W}_a \Sigma_{\phi_a, \phi_d} \mathbf{W}_d^\top \\ &= \sum_{a=1}^c \sum_{b=1}^c (\mathbf{W}_a \Sigma_\phi^{(a,b)} \mathbf{W}_b^\top) \end{aligned}$$

Use the fact that:

$$\begin{aligned} \text{Tr}(\mathbf{W}_a \mathbf{M} \mathbf{W}_b^\top) &= \text{Tr}(\mathbf{F}^* \text{diag}(\hat{\mathbf{W}}_a) \mathbf{F} \mathbf{M} \mathbf{F}^* \text{diag}(\hat{\mathbf{W}}_b)^H \mathbf{F}) \\ &= \text{Tr}(\text{diag}(\hat{\mathbf{W}}_a) \mathbf{F} \mathbf{M} \mathbf{F}^* \text{diag}(\hat{\mathbf{W}}_b)^H) \end{aligned}$$

then let $\mathbf{P}^{(a,b)} = \mathbf{F} \Sigma_\phi^{(a,b)} \mathbf{F}^*$, so $\mathbf{P}_{kk}^{(a,b)} = f_k^H \Sigma_\phi^{(a,b)} f_k$ where f_k is the k -th column of $\mathbf{F}^*$, or the conjugate of the k -th row of $\mathbf{F}$. Note that $\Sigma_\phi^{(a,b)}$ is not symmetric; rather Σ_ϕ is symmetric so $(\Sigma_\phi^{(a,b)})^\top = \Sigma_\phi^{(b,a)}$. So here we can't apply the reasoning from the linear case for $a \neq b$ to make statements about $\mathbf{P}_{kk}^{(a,b)}$, so we separate the pair indices into (a, b) for $a \neq b$ and (a, a) . For (a, a) indices, $\mathbf{P}_{kk}^{(a,a)}$ is real and non-negative, and $\mathbf{P}_{kk}^{(a,a)} = \mathbf{P}_{d-k, d-k}^{(a,a)}$. For (a, b) , we have:

$$(\mathbf{P}_{kk}^{(a,b)})^H = f_k^H (\Sigma_\phi^{(a,b)})^H (f_k^H)^H = f_k^H \Sigma_\phi^{(b,a)} f_k = \mathbf{P}_{kk}^{(b,a)}$$

and since $f_{d-k} = \overline{f_k}$:

$$\begin{aligned} \mathbf{P}_{d-k, d-k}^{(a,b)} &= f_{d-k}^H \Sigma_\phi^{(a,b)} f_{d-k} \\ &= (\overline{f_k})^H \Sigma_\phi^{(a,b)} \overline{f_k} = f_k^\top \Sigma_\phi^{(a,b)} \overline{f_k} \\ &= \text{conj}(f_k^H \Sigma_\phi^{(a,b)} f_k) = \text{conj}(\mathbf{P}_{kk}^{(a,b)}) \end{aligned}$$

Consider the first term:

$$\begin{aligned} \text{Tr}(\mathbf{W}_\sigma \Sigma_\phi \mathbf{W}_\sigma^\top) &= \sum_{a=1}^c \sum_{b=1}^c \text{Tr}(\text{diag}(\hat{\mathbf{W}}_a) \mathbf{P}^{(a,b)} \text{diag}(\hat{\mathbf{W}}_b)^H) \\ &= \sum_{a=1}^c \sum_{b=1}^c \sum_{k=1}^d \hat{w}_k^a \mathbf{P}_{kk}^{(a,b)} \text{conj}(\hat{w}_k^b) \end{aligned}$$

Pair summand (a, b) with (b, a) for $a \neq b$, such that we can cancel out the imaginary part of $\hat{w}_k^a$ and $\hat{w}_k^b$:

$$\begin{aligned} &\hat{w}_k^a \mathbf{P}_{kk}^{(a,b)} \text{conj}(\hat{w}_k^b) + \hat{w}_k^b \mathbf{P}_{kk}^{(b,a)} \text{conj}(\hat{w}_k^a) \\ &= \mathbf{P}_{kk}^{(a,b)} (\hat{w}_k^a \text{conj}(\hat{w}_k^b)) + \mathbf{P}_{kk}^{(b,a)} \hat{w}_k^b \text{conj}(\hat{w}_k^a) \\ &= \mathbf{P}_{kk}^{(a,b)} (\hat{w}_k^a \text{conj}(\hat{w}_k^b)) + \text{conj}(\mathbf{P}_{kk}^{(a,b)}) \text{conj}(\hat{w}_k^a \text{conj}(\hat{w}_k^b)) \\ &= 2\text{Re}(\hat{w}_k^a \mathbf{P}_{kk}^{(a,b)} \text{conj}(\hat{w}_k^b)) \end{aligned}$$

Therefore:

$$\text{Tr}(\mathbf{W}_\sigma \Sigma_\phi \mathbf{W}_\sigma^\top) = \sum_{a \neq b} \sum_{k=1}^d 2\text{Re}(\hat{w}_k^a \mathbf{P}_{kk}^{(a,b)} \text{conj}(\hat{w}_k^b)) + \sum_{a=1}^c \sum_{k=1}^d \|\hat{w}_k^a\|^2 \mathbf{P}_{kk}^{(a,a)}$$

Next, we evaluate the cross-covariance term $-2\text{Tr}(\mathbf{W}_\sigma \Sigma_{\phi, \mathbf{x}_0})$. Partitioning $\Sigma_{\phi, \mathbf{x}_0}$ into c blocks of $d \times d$ along its rows, denoted as $\Sigma_{\phi_a, \mathbf{x}_0}$, then:

$$\mathbf{W}_\sigma \Sigma_{\phi, \mathbf{x}_0} = \sum_{a=1}^c \mathbf{W}_a \Sigma_{\phi_a, \mathbf{x}_0}$$

and

$$\text{Tr}(\mathbf{W}_\sigma \Sigma_{\phi, \mathbf{x}_0}) = \sum_{a=1}^c \text{Tr}(\text{diag}(\hat{\mathbf{W}}_a) \mathbf{F} \Sigma_{\phi_a, \mathbf{x}_0} \mathbf{F}^*)$$

Let $\mathbf{Q}^{(a)} = \mathbf{F}\Sigma_{\phi_a, \mathbf{x}_0}\mathbf{F}^*$ with diagonal entries $Q_{kk}^{(a)} = f_k^H \Sigma_{\phi_a, \mathbf{x}_0} f_k$. Then:

$$\text{Tr}(\mathbf{W}_\sigma \Sigma_{\phi, \mathbf{x}_0}) = \sum_{a=1}^c \sum_{k=1}^d \hat{w}_k^a Q_{kk}^{(a)}$$

Since $\Sigma_{\phi_a, \mathbf{x}_0}$ has real entries, $Q_{d-k, d-k}^{(a)} = \text{conj}(Q_{kk}^{(a)})$. Also since $\mathbf{W}_\sigma$ has real entries, $\hat{w}_j^a = \text{conj}(\hat{w}_{d-j}^a)$, pairing k and $d-k$ gives:

$$\hat{w}_k^a Q_{kk}^{(a)} + \hat{w}_{d-k}^a Q_{d-k, d-k}^{(a)} = 2\text{Re}(\hat{w}_k^a Q_{kk}^{(a)})$$

Plugging back into the loss expressions:

$$\mathcal{L}_\sigma^{\text{circ}} = \sum_{k=1}^d \left[\sum_{a=1}^c \|\hat{w}_k^a\|^2 \mathbf{P}_{kk}^{(a,a)} + \sum_{a \neq b}^c 2\text{Re}(\hat{w}_k^a \mathbf{P}_{kk}^{(a,b)} \text{conj}(\hat{w}_k^b)) - 2 \sum_{a=1}^c \text{Re}(\hat{w}_k^a Q_{kk}^{(a)}) \right] + \text{Tr}(\Sigma_{\mathbf{p}_0})$$

To optimize, we group the complex weights of all c blocks into a vector $\hat{\mathbf{w}}_k = [\hat{w}_k^1, \dots, \hat{w}_k^c]^\top \in \mathbb{C}^c$. Define the Hermitian matrix $\mathbf{P}_k \in \mathbb{C}^{c \times c}$ where $(\mathbf{P}_k)_{ab} = \mathbf{P}_{kk}^{(a,b)}$, and the vector $\mathbf{q}_k \in \mathbb{C}^c$ where $(\mathbf{q}_k)_a = Q_{kk}^{(a)}$. Let $\mathbf{v}_k := \text{conj}(\hat{\mathbf{w}}_k)$. Then $\sum_{a \neq b}^c \hat{w}_k^a \mathbf{P}_{kk}^{(a,b)} \text{conj}(\hat{w}_k^b) = \mathbf{v}_k^H \mathbf{P}_k \mathbf{v}_k$ and $\sum_{a=1}^c \hat{w}_k^a Q_{kk}^{(a)} = \mathbf{v}_k^H \mathbf{q}_k$. Similar to above it looks like we can optimize w.r.t to each index k independently, however note that there is one constraint $\hat{\mathbf{w}}_k = \text{conj}(\hat{\mathbf{w}}_{d-k})$, but since $\mathbf{P}_k = \text{conj}(\mathbf{P}_{d-k})$, optimizing for both independently is still valid since the two independent solution will automatically satisfy this constraint. The objective for a given k is the quadratic form:

$$\begin{aligned} \mathcal{L}_{\sigma, k}^{\text{circ}} &= \mathbf{v}_k^H \mathbf{P}_k \mathbf{v}_k - 2\text{Re}(\mathbf{v}_k^H \mathbf{q}_k) \\ &= \mathbf{v}_k^H \mathbf{P}_k \mathbf{v}_k - \mathbf{v}_k^H \mathbf{q}_k - \mathbf{q}_k^H \mathbf{v}_k \end{aligned}$$

Taking the Wirtinger derivative w.r.t. $\mathbf{v}_k^H$ and setting it to 0:

$$\mathbf{P}_k \mathbf{v}_k - \mathbf{q}_k = 0$$

so

$$\mathbf{v}_k^* = \mathbf{P}_k^{-1} \mathbf{q}_k$$

Plugging back to get the loss:

$$\mathcal{L}_\sigma^{\text{circ}} = \text{Tr}(\Sigma_{\mathbf{p}_0}) - \sum_{k=1}^d \mathbf{q}_k^H \mathbf{P}_k^{-1} \mathbf{q}_k$$

Now it might be misleading to try to compare $\mathcal{L}_\sigma^{\text{circ}}$ with $\mathcal{L}_\sigma = \text{Tr}(\Sigma_{\mathbf{p}_0} - \text{Cov}(\mathbf{x}_0, \phi) \Sigma_\phi^{-1} \text{Cov}(\phi, \mathbf{x}_0))$ since $\mathcal{L}_\sigma$ is the unconstrained denoiser with dense RF Θ inside ϕ , however recall that Θ^{circ} is not a subset of Θ^{dense} since the joint distributions of the rows are different: rows of Θ^{circ} are dependent while those of Θ^{dense} are independent. Thus we can no longer make a blanket claim that $\mathcal{L}_\sigma^{\text{circ}} \geq \mathcal{L}_\sigma$ always, even though the $\mathbf{W}_\sigma^{\text{circ}} \subset \mathbf{W}_\sigma$ still. The point is that given the same Θ , since $\mathbf{W}_\sigma^{\text{circ}}$ and $\mathbf{W}_\sigma$ are trainable, the loss for the denoiser with the constrained $\mathbf{W}_\sigma^{\text{circ}}$ can never be more optimal than that of the unconstrained, optimal $\mathbf{W}_\sigma$. However Θ in our circulant setting and Θ in the regular unconstrained nonlinear denoiser is not the same; thus leaving space for $\mathcal{L}_\sigma^{\text{circ}}$ to beat $\mathcal{L}_\sigma$.

5 WHEN DOES $\mathcal{L}_\sigma^{\text{CIRC}}$ BEAT $\mathcal{L}_\sigma$ IN NONLINEAR DENOISER?

Through our experiments the RF denoiser needs a lot of features, i.e. $k/d \rightarrow \infty$. We know that circulant structure gives its best loss when the data is stationary, for example in a face dataset, the nose can equally be in the pixel at the top of the image as the pixel in the middle. We want to know if the circulant structure can help when we have finite k . To formalize this we first decompose the loss between the circulant structure and the dense structure:

$$\mathcal{L}^{\text{circ}}(k) - \mathcal{L}^{\text{dense}}(k) = A_{\text{circ}}(k) + \Delta_{\text{stat}}(\sigma) - A_{\text{dense}}(k)$$

where

$$A_{\text{dense}}(k) := \mathcal{L}^{\text{dense}}(k) - \text{MMSE}(\mathbf{p}_0)$$

$$A_{\text{circ}}(k) := \mathcal{L}^{\text{circ}}(k) - \text{MMSE}(\bar{\mathbf{p}}_0)$$

$$\bar{\mathbf{p}}_0(x) = \frac{1}{d} \sum_{\tau=0}^{d-1} \mathbf{p}_0(S^{-\tau} x)$$

$$\Delta_{\text{stat}}(\sigma) := \text{MMSE}(\bar{\mathbf{p}}_0) - \text{MMSE}(\mathbf{p}_0)$$

is a stationary/shift-variant distribution made from shift-symmetrizing the true data distribution i.e. $\mathbf{p}_0$, by applying a cyclic shift operator $S^{-\tau}$ to every possible shift τ , and then averaging the result.

By constraining the weights to a circulant matrix, the network is forced to be shift-equivariant. Therefore, when evaluating the non-stationary data distribution $\mathbf{p}_0$, the constrained denoiser mathematically optimizes as if the data were drawn from the shift-symmetrized (stationary) distribution $\bar{\mathbf{p}}_0$. Theoretically it can never be better than $\text{MMSE}(\bar{\mathbf{p}}_0)$. This cost is quantified by $\Delta_{\text{stat}}(\sigma)$. Note that $\bar{\mathbf{p}}_0$ is generatively $\bar{\mathbf{x}}_0 = S^T \mathbf{x}_0$, where the shift is $T \sim \text{Unif}\{0, \dots, d-1\}$. Because conditioning on T brings the problem back to the unshifted, non-stationary $\mathbf{p}_0$, the stationarization penalty is equivalent to the gap between unconditional and conditional denoising:

$$\Delta_{\text{stat}}(\sigma) = \text{MMSE}_{\bar{\mathbf{p}}_0}[\mathbf{X} | \mathbf{Y}_\sigma] - \text{MMSE}_{\bar{\mathbf{p}}_0}[\mathbf{X} | \mathbf{Y}_\sigma, T]$$

By the I-MMSE identity, integrating this penalty over the noise schedule gives the mutual information between the shifted data and the shift operator, bounded by the entropy of T :

$$\frac{1}{2} \int_0^\infty \frac{dt}{t^2} \Delta_{\text{stat}}(t) = I(\bar{\mathbf{x}}_0; T) \leq H(T) = \log d$$

Because the penalty Δ_{stat} is bounded by $\log d$ across the noise schedule, it is a finite constant independent of k . Conversely, at small k , $A_{\text{dense}}(k)$ is unconstrained/unbounded due to the $O(d^n)$ parameters required to cover the dense feature space. Furthermore, A_{circ} is a stationary problem on $\bar{\mathbf{p}}_0$, therefore we can derive this as it decomposes per-frequency like in the last section. $\mathbf{W} = 0$ is possible, so $\mathcal{L}^{\text{circ}} \leq \text{Tr}(\Sigma_{\mathbf{p}_0})$ (mean-predicting), so $0 \leq A_{\text{circ}}(k) \leq \text{Tr}(\Sigma_{\mathbf{p}_0}) - \text{MMSE}(\bar{\mathbf{p}}_0)$. If we consider the difference:

$$\mathcal{L}^{\text{circ}}(k) - \mathcal{L}^{\text{dense}}(k) = A_{\text{circ}}(k) + \Delta_{\text{stat}}(\sigma) - A_{\text{dense}}(k)$$

If $k \rightarrow 0$, both models degenerate to the bias-only predictor, $\mathcal{L}^{\text{circ}}, \mathcal{L}^{\text{dense}} \rightarrow \text{Tr}(\Sigma_{\mathbf{p}_0})$, so $A_{\text{dense}}(k) - A_{\text{circ}}(k) \rightarrow \Delta_{\text{stat}}$, so the difference $\rightarrow 0$. On the other hand if $k \rightarrow \infty$, both A vanishes, and the difference $\rightarrow \Delta_{\text{stat}} > 0$, so dense would win.

So the win region is where $A_{\text{dense}}(k) - A_{\text{circ}}(k)$ rises above its starting value Δ_{stat} before decaying to 0 along k , which can be non-monotone. Point is if we can compute these quantities from our data set then we get a set of k where the circulant denoiser outperforms the dense denoiser.